Tasfia Zaman

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Research Interests

AI safety & trustworthy AI | Mechanistic interpretability | Model interventions (LLM unlearning & knowledge editing) Representation analysis | Agent security

I am interested in when observable model behavior gives an incomplete picture of what a model represents internally, and in representation-level and causal methods for understanding, evaluating, and intervening on learned model behavior.

Education

BRAC University **Summer 2022–Fall 2025**
B.Sc. in Computer Science and Engineering *Dhaka, Bangladesh*

CGPA: 3.99/4.00 | VC's List in 10 of 11 semesters | 75% merit-based tuition scholarship

Research Experience

Undergraduate Thesis Research — Entity-Level LLM Unlearning **Feb 2025–Jan 2026**
BRAC University

- Co-developed ERUF, a representation-aware entity-level LLM unlearning framework that mines subject-specific activation signatures, suppresses signature-aligned internal activations, and distills the resulting behavior into a LoRA adapter.
- Developed and evaluated surface- and representation-level diagnostics to distinguish output suppression from internal attenuation, including SMR/EL10, adversarial and name-agnostic recovery, hidden-state probes, and sequential unlearning analyses.
- Equal-contribution work published in **Findings of EMNLP 2026**.

Knowledge Editing Research Collaboration **Feb–Jun 2026**
Remote research collaboration on associative-memory-based language-model editing

- Developed experimental and evaluation infrastructure for DSDM-based replacement and knowledge-editing studies, including data preparation, seen/unseen evaluation, run metadata, timing, and reproducibility tooling.
- Built ROME-aligned efficacy, paraphrase, and locality evaluation together with consecutive-editing workflows and bounded experiment sweeps.
- Public research artifact: [GitHub].

Mechanistic Interpretability of Attention Sinks **2026**
BlackboxNLP reproducibility and mechanism analysis

- Contributed to a collaborative reproduction and refinement of a proposed mechanistic account of transformer attention sinks using controlled representation-level and causal analyses across model settings.

Publications

Representation-Aware Unlearning via Activation Signatures: From Suppression to Entity-Signature Attenuation.
Syed Naveed Mahmood, Md. Rezaur Rahman Bhuiyan[†], **Tasfia Zaman**[†], Jareen Tasneem Khondaker, Md. Sameer Sakib, K. M. Shadman Wadith, Nazia Tasnim, Farig Sadeque.
Findings of the Association for Computational Linguistics: EMNLP 2026, pp. 11066–11087.
[ACL Anthology] [arXiv] [DOI]
[†]Equal contribution.

Same Sink, Different Plumbing: Reproducing and Refining the Mechanism of Attention Sinks.
Abdullah Al Zubayer, Syed Naveed Mahmood, Md. Rezaur Rahman Bhuiyan, Mir Md Mohiuddin Abrar, Jareen Tasneem Khondaker, **Tasfia Zaman**, Farig Sadeque.
Proceedings of the 9th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP, 2026, pp. 1233–1247.
[ACL Anthology] [DOI]

Ongoing Research

CAPE — Causally Audited Physics Editing

Post-training intervention for robust neural cosmological inference

- Studying whether post-training interventions can reduce cosmological inference sensitivity to uncertain astrophysical feedback without sacrificing cosmologically useful information.

Mind or Mouth? — Cross-Model KV Communication

Information and behavior in cross-model representation transfer

- Studying whether translated KV caches can transfer source-specific information between language models independently of learned behavioral tendencies.

MIRAGE-Persist

Security of long-running, tool-using AI systems

- Studying how security failures persist across long-running, tool-using AI systems and how to evaluate them reliably.

Selected Research & Technical Projects

HyperCite [Live] [BRACU news]

Hybrid retrieval, literature-grounded generation, and source-verifiable citations

- Built a hybrid RAG pipeline combining MongoDB Atlas vector search with lexical fallback and cross-paper evidence diversification.
- Implemented excerpt-anchored citations with page-level offsets for tracing generated claims to exact PDF locations, with real-time collaborative annotation through Liveblocks.
- Named **Champion** among 120+ projects at the BRACU CSE471 System Analysis and Design Project Showcase.

LD-Aware Genomic Prediction [GitHub]

Deep genomic prediction with LD-aware SNP representations

- Implemented and evaluated an EBMGP-style genomic-prediction pipeline using LD-aware SNP tokenization, BERT-style embeddings, convolutional regression, and multiple pooling strategies.
- Evaluated 11 quantitative traits across rice, sorghum, and Holstein-bull datasets with 5-fold cross-validation and classical regression baselines.

Teaching & Academic Experience

BRAC University **Jun 2026–Present**

Adjunct Lecturer, Department of Computer Science and Engineering *Dhaka, Bangladesh*

- Taught four undergraduate laboratory sections in Summer 2026: CSE110 (Programming Language I), CSE321 (Operating Systems), and two sections of CSE330 (Numerical Methods).
- Held weekly consultation sessions supporting students with laboratory coursework, programming concepts, and problem solving.

BRAC University **Nov 2024–May 2025**

Student Tutor *Dhaka, Bangladesh*

- Provided laboratory and one-to-one academic support for first- and second-year CSE courses, including CSE220 (Data Structures).

Selected Honors & Awards

Champion, CSE471 System Analysis and Design Project Showcase, 120+ projects	2025
2nd Place, Undergraduate Thesis Poster Presentation, 150+ teams	2025
First Runner-Up, Next Gen Ideathon, Bioinformatics Track	2025
VC's List, 10 of 11 semesters	2022–2025
75% Merit-Based Tuition Scholarship	2023–2025

Technical Skills

Research & ML: PyTorch, Hugging Face Transformers, JAX, NumPy, Pandas; representation analysis, causal intervention, LLM-unlearning and knowledge-editing evaluation

Research Computing: Python, Linux, Git, Docker, Slurm, Jupyter, LaTeX

Additional: C/C++, Java, JavaScript/TypeScript, SQL, MongoDB, MySQL/MariaDB

English: IELTS Academic 8.5 overall (Listening 9, Reading 8.5, Writing 8, Speaking 8)